

Fitra Ramadhani

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Education

HASANUDDIN UNIVERSITY

Bachelor of Electrical Engineering, Telecommunication Engineering. GPA 3.77/4

Thesis: Design of a Dual-Band Rectangular Microstrip Antenna 2.4/5.8 GHz for Wearable Radar-Based Heartbeat Monitoring Using Polyester Substrate.

Relevant Coursework: Antenna and Propagation, Electromagnetics, Transmission Lines and High Frequency, Wireless Communication Systems, and Analog Digital Communication Systems, Satellite Communication.

Makassar, Indonesia
March 2026

BANDUNG INSTITUTE OF TECHNOLOGY (ITB)

Student Exchange Program (Kemendikbud Batch 3)

Bandung, Indonesia
Aug 2023 – Dec 2023

SMK NEGERI 1 MAROS

Vocational High School - Computer and Network Engineering

Maros, Indonesia
May 2022

Experience

ANTENNA AND WAVE PROPAGATION LABORATORY

Research Staff

- Designed and optimized dual-band (2.4/5.8 GHz) microstrip antennas on flexible polyester substrates for biomedical radar applications.
- Utilized CST Studio Suite and HFSS for advanced electromagnetic simulations, focusing on Return Loss (S11), Gain, and Radiation Patterns.
- Performed physical measurements and signal accuracy analysis using a Vector Network Analyzer (VNA) to validate simulation models.

Gowa, Indonesia
February 2025 – March 2026

AIRNAV INDONESIA CABANG MATSC

Engineering Intern

- Supported the maintenance and technical troubleshooting of critical CNS (Communication, Navigation, and Surveillance) systems, including VOR, DME, and ILS.
- Assisted senior engineers in automating air traffic communication systems and conducting regular system health checks.
- Documented preventive maintenance activities and technical reports to ensure operational continuity and aviation safety standards.

Maros, Indonesia
December 2024 – February 2025

PT TRANS NASIONAL TEKNOLOGI (TRANSNET)

Engineering Intern

- Configured and maintained network infrastructure including active devices such as routers, switches, and access points.
- Executed fiber optic installation tasks, including precision splicing and connection testing for various client sites.
- Developed comprehensive site survey reports and detailed network topology documentation for infrastructure projects.

Makassar, Indonesia
June 2021 – October 2021

Leadership & Activities

HASANUDDIN UNIVERSITY ELECTRICAL ENGINEERING STUDENT ASSOCIATION

Community Service Staff Member

- Executed "Electro Mengabdi," the department's flagship community service

Gowa, Indonesia
December 2024 – May 2025

program, focusing on delivering impactful engineering-based solutions.

- Organized and implemented social service initiatives and activities tailored to address specific local community needs.
- Conducted on-site surveys and technical feasibility assessments to identify and select partner communities for future outreach programs.
- Collaborated with inter-departmental staff to ensure streamlined planning and successful implementation of community initiatives.

HASANUDDIN TECHNOFEST VIII

Member of Equipment Division

- Managed technical setup and logistical support for multiple competition categories, including robotics and engineering arenas.
- Collaborated with cross-functional teams to resolve equipment-related challenges during high-pressure event operations.

Gowa, Indonesia
March 2024 – April 2025

TELECOMMUNICATION SYSTEMS PRACTICUM, HASANUDDIN UNIVERSITY

Assistant Coordinator

- Led a team of laboratory assistants to deliver technical practicum modules for 20+ undergraduate students.
- Managed administrative workflows, including shift scheduling, task distribution, and inter-departmental coordination.
- Evaluated student performance through objective assessment of technical reports and practical laboratory examinations.

Makassar, Indonesia
January 2026 – March 2026

Project Experience

IoT-Based Automated Irrigation System

Team Project Coordinator

- Developed an end-to-end automated system using ESP32 microcontrollers and C/C++ programming.
- Integrated real-time environmental sensors (Soil Moisture, DHT21) to monitor soil and air conditions.
- Implemented Wi-Fi and TCP/IP networking protocols to transmit sensor data to the Blynk Cloud platform.
- Designed a cloud-based architecture for remote monitoring and data-driven irrigation control.

Makassar, Indonesia
March 2025 – January 2026

Training and Certification

Fundamental of Drivetest Technician - National

Digital Talent Academy (Komdigi)

Credential ID: 21212117840-783/DTA/BLSDM.Komdigi/2026

- Certified in cellular network optimization, RF parameter analysis (RSRP, SINR), and standardized field data collection procedures.

Online
May 2026

Advanced Routing, Infrastructure, and Network Management

Packt (via Coursera)

Credential ID: coursera.org/verify/M784YF9Q5X8B

- Advanced training in OSPF routing, VLAN optimization, and network performance monitoring for enterprise-scale environments.

Online
Jan 2025

Safety in the Utility Industry

University at Buffalo (via Coursera)

Credential ID: coursera.org/verify/YWB43FH0N5CM

- Trained in HSE standards for utility infrastructure, including hazard identification

Online
Feb 2025

(HazCom), OSHA regulations, and equipment safety protocols.

Introduction to Computers

Microsoft (via Coursera)

Credential ID: coursera.org/verify/Z6P2R9T4L7WQ

Online

Feb 2025

- Gained foundational expertise in computer architecture, enterprise IT systems, and operating system management.

Certificate of Competency: Computer and Network Engineering

Universitas Dipa Makassar

Credential ID: 036 / UNDIPA / B.1 / V / 2022

Maros, Indonesia

May 2022

- National recognition for technical skills in installation, configuration, and troubleshooting of active network devices.

Skills & Interests

Technical: Antenna Design (CST Studio Suite), RF Engineering, Python, C/C++, IoT Development (ESP32), TCP/IP Networking, Routing & Switching, Drivetest Engineering.

Language: Bahasa Indonesia (Native), English (Intermediate).

Software: CST Studio Suite, AWR Design Environment, Cisco Packet Tracer, Nemo Outdoor, Word, Power Point, MATLAB.

Interests: Outdoor activities, Strategy Gaming, Analytical Philosophy, Technology Exploration